Chem 220L – Measuring Manganese in Steel by Spectrophotometry with Standard Addition

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Abstract

The manganese content of an unknown steel sample ([unknown no.]) was determined by visible spectrophotometry using the method of standard addition. Three ∼0.6 g portions of the steel were dissolved in hot 3 M HNO_3, treated with ammonium persulfate to oxidize carbon, filtered, and diluted to 250 mL. A 25.00 mL aliquot of each solution was treated with H_3PO_4 to mask Fe^{3+} and with KIO_4 to oxidize Mn^{2+} to violet MnO_4^-; a second aliquot received in addition 0.500 mg of standard Mn^{2+}. Absorbances were measured at 525 nm against a reagent blank. The concentration of added manganese in the 50 mL measurement flask was 10.00 µg mL^{-1} (1.820×10^{-4} M), and the permanganate concentrations in the unknown flasks were 7.148×10^{-5}, 8.566×10^{-5}, and 7.764×10^{-5} M. These correspond to 0.3258 %, 0.3902 %, and 0.3585 % manganese by mass (mean 0.3582 %, $s = 0.0322$ %; percent relative range 18.0 %). An uncertainty analysis shows that measurement uncertainty (≈ 1 % relative) is far too small to explain the replicate spread, so the disagreement most likely originates mainly in sample dissolution, sampling, or sample handling. One known error, however, was not captured in that estimate: the standard-addition flask for replicate B was accidentally filled above the volumetric mark. This dilution lowers A_{S+X} for B and biases its result high, so the 0.3902 % value for B should be regarded as an upper limit (Section 6.3).

1 Introduction

1.1 Why Manganese Is Added to Steel

Steel is an alloy of iron that typically contains about 0.5 % manganese by mass, together with carbon and many other elements. Manganese is added deliberately for several reasons. Most importantly, it combines with sulfur, an unavoidable impurity from iron ore and coke, to form manganese sulfide (MnS) in preference to iron sulfide (FeS). Iron sulfide forms a low-melting film along grain boundaries; when the steel is hot-rolled or forged, this film melts and the metal cracks ("hot shortness"). Manganese sulfide has a much higher melting point and forms discrete inclusions, so the steel remains workable at forging temperatures. Manganese also acts as a deoxidizer during steelmaking by combining with dissolved oxygen, and it dissolves in the iron lattice, increasing strength, hardness, and hardenability. Because the amount of manganese controls these properties, its concentration is a routine specification in steel quality control.

1.2 How an Absorbance Spectrophotometer Works

A UV–visible absorption spectrophotometer measures how much light a sample removes from a beam of known wavelength. A lamp (commonly a tungsten-halogen lamp for the visible region and a deuterium lamp for the ultraviolet) provides broadband light. A monochromator, which uses a diffraction grating or prism and a narrow slit, selects a narrow band of wavelengths, here 525 nm. The selected light passes through a cuvette of known path length b (1.000 cm) containing the sample, and a detector (a photodiode, photomultiplier tube, or detector array) measures the transmitted radiant power P. Because the cuvette walls, the solvent, and any colored reagents also absorb or reflect light, the instrument first measures the power P_0 through a reference or blank solution and sets this to zero absorbance ("autozero"). Absorbance is then

$$A = -\log_{10}\left(\frac{P}{P_0}\right) = \varepsilon\, b\, c, \tag{1}$$

where ε is the molar absorptivity and c is the concentration of the absorbing species. The second equality is the Beer–Lambert law; it holds for dilute solutions of a single absorber measured with nearly monochromatic light, and fails at high concentration, when the absorber chemically changes, or when stray light or scattering is significant. The logarithmic scale is why absorbance is the useful quantity: it is directly proportional to concentration, whereas transmittance is not. Because the absorbance of a solution depends on everything in it, only species that absorb at the chosen wavelength matter, and the reagent-blank subtraction removes the background from reagents and cuvettes.

Manganese(II) is nearly colorless, so it is first converted to permanganate, which has an intense violet color with a strong absorption band near 525 nm:

$$2\,\mathrm{Mn^{2+}(aq)} + 5\,\mathrm{IO_4^-(aq)} + 3\,\mathrm{H_2O(l)} \longrightarrow 2\,\mathrm{MnO_4^-(aq)} + 5\,\mathrm{IO_3^-(aq)} + 6\,\mathrm{H^+(aq)}. \tag{2}$$

1.3 The Standard Addition Technique

An ordinary external calibration curve is made from standards prepared in clean solvent. It gives the wrong answer if other components of the sample (the *matrix*) change the instrument response to the analyte, for example by absorbing, scattering, or reacting with it. Steel digests contain a large excess of iron and many other elements, so a matrix effect is plausible. In the method of *standard addition*, a known amount of analyte is added directly to a portion of the sample, so the added analyte experiences the same matrix as the original analyte. The unknown is measured twice: once as prepared (signal I_X) and once after the standard has been added (signal I_{S+X}). The increase in signal is produced by a known quantity of analyte, which calibrates the response in the real matrix. If the unknown aliquot of concentration $[X]_i$ is diluted to the same final volume in both flasks, then $[X]_f = [X]_i$ and the signals are in the ratio

$$\frac{[X]_i}{[S]_f + [X]_f} = \frac{I_X}{I_{S+X}} \quad\Longrightarrow\quad [X]_f = [S]_f\,\frac{I_X}{I_{S+X} - I_X}, \tag{3}$$

where $[S]_f$ is the concentration of added standard in the final flask (Harris, Eq. 5.8). Graphically, a straight line through the two points (added concentration, signal) crosses zero signal at a negative added concentration equal to $-[X]_f$ (Figure 3). The technique assumes a linear response and a signal that is zero when no analyte is present, which is why the reagent blank is subtracted.

2 Safety

Goggles, a lab coat, and gloves were worn. Safety data sheets were reviewed for acetone, nitric acid, ammonium persulfate, ammonium hydrogen sulfite, potassium periodate, and phosphoric acid. Boiling nitric acid releases nitrogen oxides, so the digestions were done in the fume hood under watch glasses, and the solutions were not allowed to boil dry. Acetone is flammable and was kept away from the hot plate. Hot glassware was handled with care. Organic waste (acetone) and permanganate-containing waste were collected in their designated containers.

3 Methods

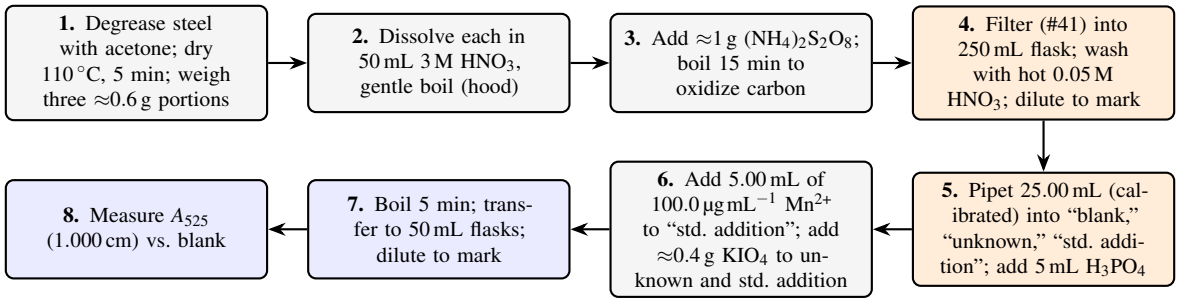


Figure 1: Workflow for the manganese determination (performed for three replicate steel portions).

3.1 Sample Dissolution (September 30)

About 1.8 g of the steel (recorded: 1.8002 g) was rinsed in a weighing bottle with about 5 mL of acetone to remove oil and grease. The acetone was decanted into the organic waste container, the steel was dried at 110 °C for 5 min, and it was cooled in a desiccator. Three portions of about 0.6 g were weighed to 0.1 mg into separate 250 mL beakers (A, B, C). About 300 mL of 3 M HNO_3 was prepared from concentrated 15.8 M HNO_3. Each sample was dissolved in 50 mL of 3 M HNO_3 by gently boiling under a watch glass in the hood (stopping after 1 h if particles remained and replacing evaporated acid, without allowing the solution to evaporate completely). After cooling for at least 5 min, about 1 g of ammonium persulfate was added to each beaker and the solution was boiled for 15 min to oxidize carbon to CO_2:

$$\mathrm{C(s) + 2\,S_2O_8{}^{2-}(aq) + 2\,H_2O(l) \longrightarrow CO_2(g) + 4\,SO_4{}^{2-}(aq) + 4\,H^+(aq).} \tag{4}$$

No pink permanganate or brown MnO_2 persisted after the persulfate treatment, so the NH_4HSO_3 reduction step was not needed.

3.2 Filtration (October 5)

Each solution was cooled to near room temperature and filtered quantitatively through #41 filter paper into a 250 mL volumetric flask. The beaker was washed many times with small portions of hot (not boiling) 0.05 M HNO_3, which were passed through the filter. After cooling, each flask was diluted to the mark with purified water and mixed. About 100 mL of each solution was set aside (flasks A, B, C) for a later lab.

3.3 Calibration of the 25 mL Pipet

The mass of a clean, dry stoppered 125 mL Erlenmeyer flask was measured. Water was delivered from the 25.00 mL pipet into the flask, which was stoppered and reweighed; the procedure was repeated three more times without re-drying the flask (four deliveries in all). The mass of water delivered was converted to volume using the density table in the textbook (Harris, Table 2-7). The average volume was used for all calculations that involved this pipet.

3.4 Preparation of the Spectrophotometric Samples (October 7)

For each of A, B, and C, 25.00 mL (calibrated pipet) was transferred into three 100 mL beakers labeled "blank," "unknown," and "standard addition," and 5 mL of 85 wt% H_3PO_4 was added to each. 5.00 mL of the instructor's standard Mn^{2+} ($100.0(5)\ \mu g\,mL^{-1}$ in 0.05 M HNO_3) was added to the "standard addition" beaker, and about 0.4 g of KIO_4 was added to the "unknown" and "standard addition" beakers. These six beakers were boiled gently for 5 min until the KIO_4 dissolved. All nine beakers were then transferred quantitatively (washing the beaker many times) into 50 mL volumetric flasks, diluted to the mark, and mixed by inversion. **For replicate B, distilled water was added past the mark of the standard-addition flask, so its final volume was larger than** 50.00 mL**.** The overfilled volume was not measured and the solution was not remade; the consequences are analyzed in Section 6.3.

3.5 Spectrophotometry

The nine solutions were measured at 525 nm in matched 1.000 cm cuvettes, rinsing each cuvette with the solution to be measured. The spectrophotometer was zeroed with the blank in the reference position, and the absorbance of the unknown and standard-addition solutions was recorded for each replicate. The blank reading recorded for each replicate was also subtracted, as described in Section 5.

4 Results

4.1 Raw Data

The masses of the three steel portions and the measured absorbances are given in Table 1. Standard addition raised the absorbance of each solution by about 0.44–0.49 absorbance units, close to the 0.45 that the lab manual predicts for 0.50 mg of added manganese.

Table 1: Absorbances (525 nm) of unknown ([unknown no.]) and standard addition for each sample.

		Absorbance (*A*)		
Entry	**Mass (g)**	**Blank**	**Unknown**	**Standard addition**
A	0.6023	0.002	0.174	0.612
B	0.6026	0.002	0.210	0.652
C	0.5945	0.004	0.213	0.703

4.2 Pipet Calibration

The 25.00-mL pipet delivered a mean volume of 25.014 mL (Table 2), 0.06 % above its nominal value; this volume, $V_{\text{pip}} = 25.014\,\text{mL}$, was used in all later calculations. The water temperature was not recorded, so a density factor for about 22 °C (1.0033 mL g^{-1}, which includes the air-buoyancy correction) was used. Using the 20 °C or 25 °C factors would change the volume by less than 0.07 % and the final results by less than 0.0003 percentage points.

Table 2: Calibration of the 25.00 mL pipet (mass of water delivered, converted with 1.0033 mL g^{-1}).

Trial	Flask + water (g)	Water delivered (g)	Volume (mL)
	94.5841 (empty flask + stopper)	–	–
1	119.4863	24.9022	24.984
2	144.4155	24.9292	25.012
3	169.3558	24.9403	25.023
4	194.3126	24.9568	25.039
		Mean ± SD	25.014 ± 0.023

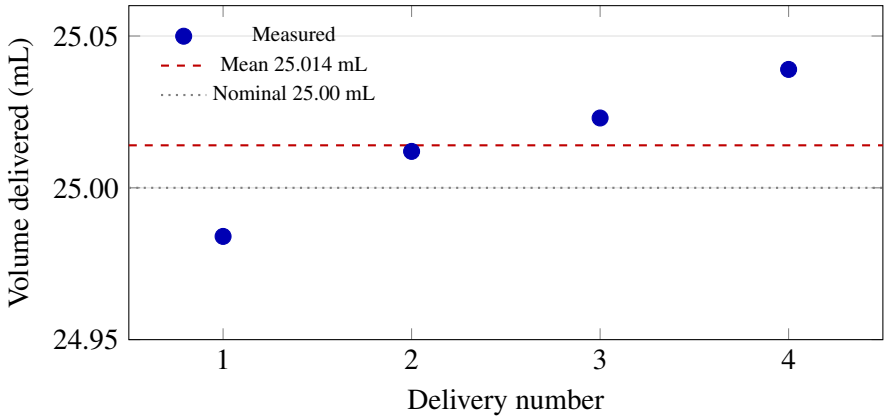


Figure 2: Pipet calibration. The delivered volume increased from the first to the fourth delivery; the pipet was probably draining more completely as the technique settled.

4.3 Concentration of Added Manganese

The 5.00 mL of 100.0 µg mL^{-1} standard was diluted to 50.00 mL, so the concentration of added manganese in the measurement flask was

$$[S]_f = \frac{(100.0\,\mu\text{g mL}^{-1})(5.00\,\text{mL})}{50.00\,\text{mL}} = 10.00\,\mu\text{g mL}^{-1} = \boxed{1.820 \times 10^{-4}\,\text{M}}.$$

4.4 Permanganate Concentration and Weight Percent Manganese

Table 3 gives the blank-corrected absorbances, the permanganate concentration $[X]_f$ in each 50 mL unknown flask (Equation (3) with $[X]_f = [X]_i$), and the weight percent manganese in the steel. The concentrations of MnO_4^- in the unknown flasks were 7.148×10^{-5} M (A), 8.566×10^{-5} M (B), and 7.764×10^{-5} M (C), assuming all Mn^{2+} was oxidized. The weight percent manganese was 0.3258 % (A), 0.3902 % (B), and 0.3585 % (C), with a mean of 0.3582 %. These values assume that all three standard-addition flasks had a final volume of exactly 50.00 mL. That assumption is not valid for B, whose flask was overfilled (Section 6.3), so the result for B is biased high by an unknown amount.

Table 3: Blank-corrected absorbances, permanganate concentrations, and manganese content of unknown [unknown no.].

Entry	A_X	A_{S+X}	ΔA	$[X]_f$ **(M)**	$[X]_f$ ($\mu g\,mL^{-1}$)	**Mn (mg, 250 mL)**	**wt% Mn**
A	0.172	0.610	0.438	7.148×10^{-5}	3.927	1.962	0.3258
B	0.208	0.650	0.442	8.566×10^{-5}	4.706	2.352	0.3902
C	0.209	0.699	0.490	7.764×10^{-5}	4.265	2.131	0.3585
						Mean	0.3582
						Standard deviation	0.0322
						%RSD	8.99
						95% CI ($t = 4.303$, $n = 3$)	± 0.0800

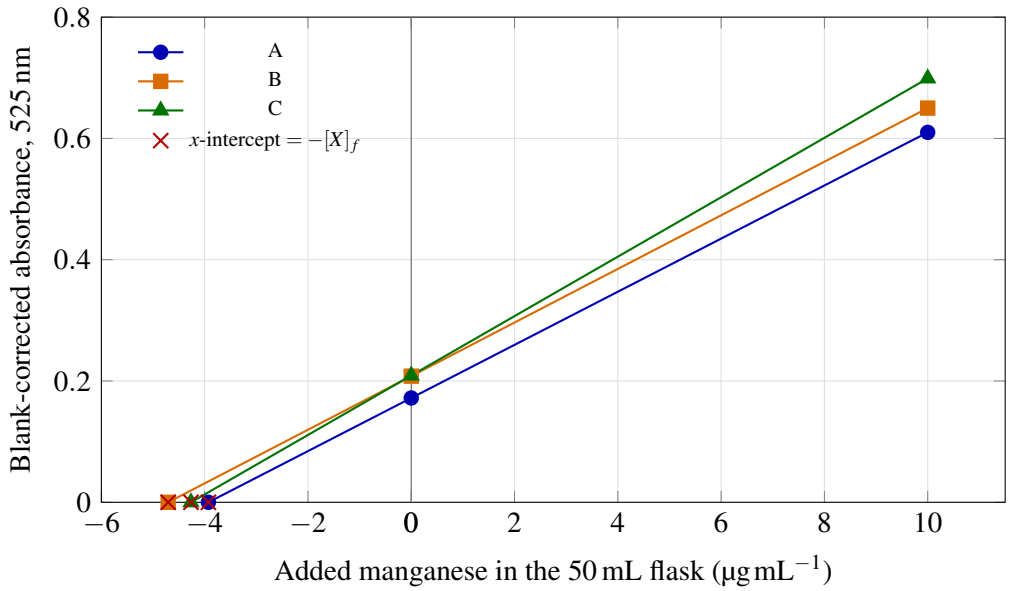


Figure 3: Standard-addition plots. The solid points are the measured unknown (0 added) and standard-addition (10.00 $\mu g\,mL^{-1}$ added) absorbances; the line is extrapolated to zero absorbance, and the x-intercept (red $\times$) is $-[X]_f$.

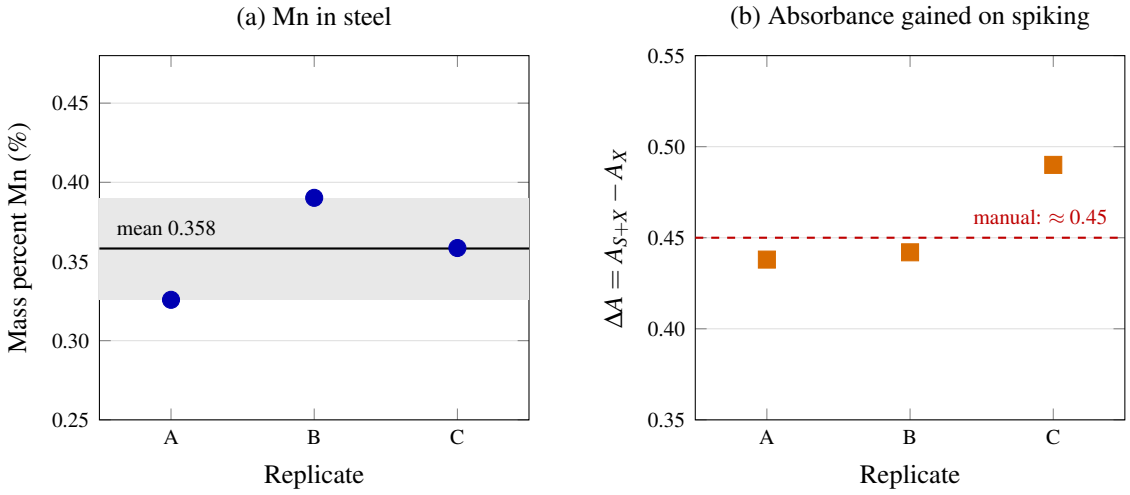


Figure 4: (a) Manganese content of the three replicates (shaded band $= \pm 1s$ about the mean). Note the truncated y-axis. (b) The absorbance increase produced by the same 0.500 mg spike; it is nearly identical for A and B but about 11 % higher for C.

5 Calculations

Calculations are shown for replicate A; B and C were treated identically.

5.1 Pipet Volume

$$m_{\text{water}} = 119.4863\,\text{g} - 94.5841\,\text{g} = 24.9022\,\text{g}, \qquad V = (24.9022\,\text{g})(1.0033\,\text{mL g}^{-1}) = 24.984\,\text{mL}.$$

The four volumes average 25.014 mL ($s = 0.023$ mL, 0.09 %).

5.2 Blank Correction

The blank absorbance recorded with each replicate (0.002 for A) was subtracted from both the unknown and the standard-addition absorbances:

$$A_X = 0.174 - 0.002 = 0.172, \qquad A_{S+X} = 0.612 - 0.002 = 0.610.$$

5.3 Permanganate Concentration

The final volume of both flasks is 50.00 mL, so $[X]_f = [X]_i$ and

$$[X]_f = [S]_f \frac{A_X}{A_{S+X} - A_X} = (1.820 \times 10^{-4}\,\text{M}) \frac{0.172}{0.610 - 0.172}$$
$$= (1.820 \times 10^{-4}\,\text{M})(0.3927) = \boxed{7.148 \times 10^{-5}\,\text{M}}\ (= 3.927\,\mu\text{g mL}^{-1}\ \text{as Mn}).$$

5.4 Weight Percent Manganese

Each permanganate ion contains one manganese atom, so the manganese in the 50 mL flask came from the 25.014 mL aliquot of the 250 mL steel solution:

$$m_{\mathrm{Mn,\,50\,mL}} = (3.927\,\mu\mathrm{g\,mL^{-1}})(50.00\,\mathrm{mL}) = 196.3\,\mu\mathrm{g},$$
$$m_{\mathrm{Mn,\,250\,mL}} = 196.3\,\mu\mathrm{g} \times \frac{250\,\mathrm{mL}}{25.014\,\mathrm{mL}} = 1962\,\mu\mathrm{g} = 1.962\,\mathrm{mg},$$
$$\%\,\mathrm{Mn} = \frac{1.962 \times 10^{-3}\,\mathrm{g}}{0.6023\,\mathrm{g}} \times 100 = \boxed{0.3258\,\%}.$$

No buoyancy correction was applied to the steel masses, as directed by the procedure.

5.5 Mean, Standard Deviation, and Percent Relative Range

$$\bar{x} = \frac{0.3258 + 0.3902 + 0.3585}{3} = \boxed{0.3582\,\%},$$
$$s = \sqrt{\frac{(0.3258-0.3582)^2 + (0.3902-0.3582)^2 + (0.3585-0.3582)^2}{3-1}} = \boxed{0.0322\,\%},$$
$$\%\mathrm{RSD} = \frac{0.0322}{0.3582} \times 100 = \boxed{8.99\,\%},$$
$$\%\text{ relative range} = 100 \times \frac{0.3902 - 0.3258}{0.3582} = \boxed{17.99\,\%}.$$

Dixon's Q for the extreme values is 0.51, below the 95% critical value of 0.970 ($n = 3$), so no replicate can be rejected. The 95% confidence interval is $\bar{x} \pm ts/\sqrt{n} = 0.3582 \pm (4.303)(0.0322)/\sqrt{3} = 0.3582 \pm 0.0800$.

5.6 Check on the Molar Absorptivity

The spike alone raises the absorbance by ΔA, so $\varepsilon = \Delta A/([S]_f\, b)$. For A, $\varepsilon = 0.438/[(1.820 \times 10^{-4}\,\mathrm{M})(1.000\,\mathrm{cm})] = 2.41 \times 10^3\,\mathrm{M^{-1}\,cm^{-1}}$; B and C give 2.43×10^3 and $2.69 \times 10^3\,\mathrm{M^{-1}\,cm^{-1}}$. These are of the magnitude generally reported for permanganate near 525 nm (about 2.3×10^3 to $2.5 \times 10^3\,\mathrm{M^{-1}\,cm^{-1}}$), which supports the assumption that essentially all of the added manganese was oxidized.

6 Discussion

6.1 Precision of the Result

The three replicates gave 0.3258 %, 0.3902 %, and 0.3585 % manganese (mean 0.3582 %). The **percent relative range is** 18.0 % and the %RSD is 9.0 %. This is a large spread for three portions of the same steel, and it is the main anomaly of this experiment. The mean of 0.358 % is lower than the roughly 0.5 % manganese said to be typical of steel, but that figure is only a typical value and cannot be used to judge the accuracy of an unknown; the true value of this sample was not known to us.

6.2 Is the Spread Caused by the Measurement?

A Monte Carlo propagation of the measurement uncertainties (Table 4) gives a relative uncertainty of only about 1 % (0.004 percentage points) in each result. The assumptions were ± 0.001 in each absorbance reading (including the blank), 0.5 % in the standard concentration, 0.2 % in the 5.00 mL pipet, 0.1 mg in each steel mass, and the observed 0.09 % scatter of the calibrated pipet; the 250 mL flask was assumed to be accurate to 0.05 %. The observed standard deviation (0.032 %) is about eight times larger. The spread therefore cannot be explained by random uncertainty in the balance, pipets, standard, or spectrophotometer. The overfilled flask for B is a separate, systematic error that is not included in this estimate and is treated below; apart from that, most of the spread must come from differences between the replicate samples before the measurement step.

Table 4: Sensitivity of the results to data-handling choices (wt% Mn).

Treatment	**A**	**B**	**C**	**Mean**
Reported (blank-corrected, $V_{\text{pip}} = 25.014$ mL)	0.3258	0.3902	0.3585	0.3582
No blank subtraction	0.3296	0.3940	0.3654	0.3630
Nominal pipet volume (25.00 mL)	0.3260	0.3905	0.3587	0.3584
Replace each ΔA by the manual's 0.450	0.3171	0.3833	0.3904	0.3636
C calculated with mean ΔA of A and B	0.3258	0.3902	0.3993	0.3718
B flask taken as 51 mL (overfilled 1 mL)	0.3258	0.3791	0.3585	0.3545
B flask taken as 52 mL (overfilled 2 mL)	0.3258	0.3686	0.3585	0.3510
B flask taken as 53 mL (overfilled 3 mL)	0.3258	0.3586	0.3585	0.3476
Estimated uncertainty (Monte Carlo, 1 s.d.)	± 0.0038	± 0.0040	± 0.0036	–

None of the alternative treatments in the first five rows removes the disagreement: the relative range stays between 18 % and 20 % in every case, and the mean changes by at most 0.014 percentage points. The conclusion is therefore not sensitive to the blank, pipet calibration, or temperature assumptions. A sharper clue comes from the unknown absorbances alone: for nearly identical steel masses (0.6023 g, 0.6026 g, 0.5945 g), A_X was 0.172, 0.208, and 0.209, so the absorbance per gram of steel differs by 20 % between replicates. This difference is present before any standard is added and must therefore arise in the dissolution, filtration, dilution, or oxidation steps, or from the steel itself.

6.3 Effect of the Overfilled Flask for Replicate B

For replicate B the standard-addition flask was filled with water above the 50 mL mark, so its final volume V' was larger than 50.00 mL. The unknown flask for B was not affected. In the overfilled flask, both the added manganese and the manganese from the steel were diluted by the factor $f = 50.00/V' < 1$, so the measured absorbance was

$$A_{S+X}^{\text{meas}} = f\,(A_X + \Delta A_{\text{true}}), \qquad \text{so} \qquad A_{S+X}^{\text{true}} = \frac{A_{S+X}^{\text{meas}}}{f} = A_{S+X}^{\text{meas}}\,\frac{V'}{50.00}.$$

Using the uncorrected A_{S+X} makes the absorbance increase from the spike look too small, and since $[X]_f = [S]_f A_X/(A_{S+X} - A_X)$, this makes the calculated manganese concentration too large. The error for B is therefore *high*, and the reported 0.3902 % is an upper limit. Because V' was not measured, the result

cannot be corrected exactly, but it can be bracketed. If the final volume was $V' = 51\,\text{mL}$, 52 mL, or 53 mL, the corrected result for B would be 0.379 %, 0.369 %, or 0.359 % (Table 4), and the percent relative range of the three replicates would fall to 15.0 %, 12.2 %, or 9.4 %.

Two observations limit how much of the spread this error can explain. First, ΔA for B (0.442) is not smaller than for A (0.438), which would be expected if the flask were much too dilute; however, ΔA for C was 11 % different from A and B for unrelated reasons, so this comparison can exclude only large overfills, and a dilution of 1–3 mL (2–6%) remains possible. Second, an overfill of about 4.8 mL would be needed for B to equal the mean of A and C, which is much more than a 50 mL flask can hold above the mark. The overfilled flask therefore probably accounts for part of B's high value, but not for all of the replicate disagreement: even with a large correction the relative range remains near 10 %, because replicate A is still low. The unknown absorbance per gram of steel was 0.286, 0.345, and 0.352 g for A, B, and C, and these values do not involve the standard-addition flasks at all; B and C agree within 2 % and A is about 18 % lower. If the volume of the overfilled flask can be estimated from the notebook or from the neck of the flask, the corrected value for B should be calculated with the equation above.

6.4 Possible Causes

- **Sample inhomogeneity.** The portions were small ($\sim$0.6 g) pieces of a metal sample. Manganese is partly present as MnS inclusions, which may not be evenly distributed, so portions of the same steel may genuinely contain different amounts. This would not be visible in the other steps.

- **Incomplete dissolution or retention on the filter.** Undissolved particles and any MnO_2 or manganese-bearing residue retained on the filter paper would lower the manganese in the flask. The notebook records that the samples turned brown on heating and green after persulfate was added, but no NH_4HSO_3 was used. A brown color is consistent with iron(III) species or with some MnO_2, and a small amount of residual MnO_2 would bias a result low. Replicate A (lowest) is the one whose result is most affected by such a loss, but the notebook does not tell us which sample lost the most.

- **Incomplete oxidation of Mn^{2+}.** Because both the unknown and spiked flasks contain KIO_4, incomplete oxidation would affect both; if it were incomplete in the unknown by the same fraction it would partly cancel in the standard-addition ratio, but any difference in boiling time between the beakers would not cancel.

- **Anomalous spike response in C.** The spike should raise every absorbance by the same amount, but ΔA for C (0.490) is about 11 % larger than for A and B (0.438, 0.442), and the apparent molar absorptivity is correspondingly higher ($2.69 \times 10^3\,\text{M}^{-1}\,\text{cm}^{-1}$ vs. 2.41×10^3 and 2.43×10^3). If A_{S+X} for C is too high, perhaps from a bubble, a fingerprint, or an unmatched cuvette, the weight percent for C would be too low: using the mean ΔA of A and B raises it from 0.3585 % to 0.3993 %. This is a possibility, not a correction, and the data do not allow us to confirm it.

- **Transcription and sampling issues in the notebook.** The mass of A was written as 0.5992 g and then corrected to 0.6023 g, and the blank for B was written as 0.018 and then corrected to 0.002. Using 0.5992 g would change A by only 0.002 percentage points, so the correction is not responsible for the spread, but the notebook entries are less certain than we would like.

6.5 Other Observations and Improvements

The pipet delivered 25.014 mL, but the volumes rose steadily with each delivery (24.984 to 25.039 mL), suggesting that the pipet was draining more completely as the technique improved; the effect on the results is 0.06 % or less. The water temperature was not recorded, but this has a negligible effect (see Table 4). The recorded blank readings were 0.002–0.004, so it is important to know whether the instrument was autozeroed against the blank (in which case the unknown and spiked absorbances were already blank-corrected); subtracting the blank a second time would lower the results by about 1 % (Table 4), which is small compared with the replicate spread. Because the spread appears to arise before the spectrophotometric step, the most useful improvements would be to dissolve larger or more finely divided portions of steel, check the filter and any residue for retained manganese, add the NH_4HSO_3 step when any brown color remains, repeat the standard-addition measurement of each flask in duplicate, and measure the spiked and unspiked solutions in the same cuvette. An overfilled volumetric flask cannot be corrected afterwards, so any flask filled past the mark should be remade (or its volume measured and recorded at once).

7 Conclusion

The manganese content of the unknown steel ([unknown no.]) was determined by standard addition after oxidation to permanganate. The concentration of added manganese in the measurement flasks was 1.820×10^{-4} M, and the permanganate concentrations in the unknown flasks were 7.148×10^{-5}, 8.566×10^{-5}, and 7.764×10^{-5} M, corresponding to 0.3258 %, 0.3902 %, and 0.3585 % manganese. The mean was 0.3582 % with $s = 0.0322$ % and a percent relative range of 18.0 %. The standard-addition absorbance increases (0.44–0.49) and the apparent molar absorptivity ($\approx 2.4 \times 10^3\,\mathrm{M^{-1}\,cm^{-1}}$ for A and B) were consistent with expectations, and the estimated measurement uncertainty (≈ 1 %) was much smaller than the replicate spread. The result of 0.358 % should therefore be regarded as a mean with a wide 95% confidence interval ($\pm$0.080 %). Part of the disagreement is explained by an overfilled standard-addition flask for replicate B, which biases that result high (a final volume of 52 mL would lower it to 0.369 %), but a large relative range would remain even after any plausible correction, pointing to differences between the digested portions as the main cause.

Waste Disposal

Acetone rinses were placed in the organic waste container. Permanganate-containing solutions were collected in the permanganate waste container in the fume hood, and nitric acid solutions in the acid waste container.

References

1. BYU–Idaho Chemistry Laboratory. *Experiment #5: Measuring Manganese in Steel by Spectrophotometry with Standard Addition*, Chem 220L.

2. Harris, D. C. *Quantitative Chemical Analysis*, 10th ed.; W. H. Freeman: New York, 2020; Chapters 2, 5, and 18.

3. Skoog, D. A.; West, D. M.; Holler, F. J.; Crouch, S. R. *Fundamentals of Analytical Chemistry*, 9th ed.; Cengage: Belmont, CA, 2014.